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SCHOOL/INSTITUTION:	Mopani park secondary school
EXAM BOARD:	ZIMSEC
SUBJECT:	CHEMISTRY
LEVEL / GRADE:	O-LEVEL
PAPER:	Paper 1 — Multiple Choice
TOTAL MARKS:	40 marks
TIME ALLOWED:	1h

ZIMSEC Chemistry O-Level — Paper 1 Mock Examination

Topic: Concentration

INSTRUCTIONS TO CANDIDATES

1. Write your name and school in the spaces provided on the front page.
2. This paper contains 40 questions. Each question carries ONE mark.
3. Read each question carefully before answering.
4. For each question, choose ONE answer: A, B, C or D.
5. Circle or clearly write the letter of your chosen answer.
6. Do NOT spend too much time on any single question.
7. If you change your answer, cross it out clearly and write your new answer.

Here is the ZIMSEC O-Level Chemistry Paper 1 (Multiple Choice) examination focusing on the topic "Concentration".

Q1. A solution contains 10g of sugar in 100cm³ of water. What is the concentration of the sugar in the solution?

- A) 0.1M
- B) 0.5M
- C) 10M
- D) 100M

Q2. What is the term for the amount of substance of a solute per unit volume of a solution?

- A) Molarity
- B) Molality
- C) Mass concentration
- D) Volume concentration

Q3. A solution contains 20g of salt in 500cm³ of water. What is the mass concentration of the salt in the solution?

- A) (20/500)g/cm³
- B) (20/100)g/cm³
- C) (500/20)g/cm³
- D) (100/20)g/cm³

Q4. What is the unit of molarity?

- A) mol/L
- B) mol/m³
- C) g/L
- D) g/m³

Q5. A solution contains 10cm³ of a 2M solution of HCl. What volume of water should be added to make a 1M solution?

- A) 50cm³
- B) 100cm³
- C) 200cm³
- D) 500cm³

Q6. A solution contains 30g of sugar in 200cm³ of water. What is the molarity of the sugar in the solution?

- A) 0.15M
- B) 0.3M
- C) 1.5M

D) 3M

Q7. What is the term for the amount of substance of a solute per unit mass of a solution?

- A) Molarity
- B) Molality
- C) Mass concentration
- D) Volume concentration

Q8. A solution contains 40g of salt in 400cm³ of water. What is the mass concentration of the salt in the solution?

- A) (40/400)g/cm³
- B) (40/100)g/cm³
- C) (400/40)g/cm³
- D) (100/40)g/cm³

Q9. What is the unit of molality?

- A) mol/kg
- B) mol/L
- C) g/kg
- D) g/L

Q10. A solution contains 20cm³ of a 3M solution of NaOH. What volume of water should be added to make a 2M solution?

- A) 50cm³
- B) 100cm³
- C) 150cm³
- D) 200cm³

Q11. A solution contains 50g of sugar in 100cm³ of water. What is the mass concentration of the sugar in the solution?

- A) (50/100)g/cm³
- B) (50/50)g/cm³
- C) (100/50)g/cm³
- D) (50/50)g/cm³

Q12. What is the term for the amount of substance of a solute per unit volume of a solution, expressed in terms of moles per unit volume?

- A) Molarity
- B) Molality
- C) Mass concentration
- D) Volume concentration

Q13. A solution contains 30cm³ of a 4M solution of H₂SO₄. What volume of water should be added to make a 2M solution?

- A) 100cm³
- B) 150cm³
- C) 200cm³
- D) 300cm³

Q14. A solution contains 60g of salt in 300cm³ of water. What is the mass concentration of the salt in the solution?

- A) (60/300)g/cm³
- B) (60/100)g/cm³
- C) (300/60)g/cm³

D) $(100/60)\text{g/cm}^3$

Q15. What is the unit of mass concentration?

- A) g/cm^3
- B) g/L
- C) mol/kg
- D) mol/L

Q16. A solution contains 40cm^3 of a 2M solution of NaCl. What volume of water should be added to make a 1M solution?

- A) 100cm^3
- B) 150cm^3
- C) 200cm^3
- D) 250cm^3

Q17. A solution contains 20g of sugar in 50cm^3 of water. What is the molarity of the sugar in the solution?

- A) 0.4M
- B) 0.8M
- C) 2M
- D) 4M

Q18. What is the term for the amount of substance of a solute per unit mass of a solution, expressed in terms of moles per unit mass?

- A) Molarity
- B) Molality
- C) Mass concentration
- D) Volume concentration

Q19. A solution contains 50cm^3 of a 3M solution of H_2SO_4 . What volume of water should be added to make a 1M solution?

- A) 100cm^3
- B) 150cm^3
- C) 200cm^3
- D) 300cm^3

Q20. A solution contains 60g of sugar in 200cm^3 of water. What is the mass concentration of the sugar in the solution?

- A) $(60/200)\text{g/cm}^3$
- B) $(60/100)\text{g/cm}^3$
- C) $(200/60)\text{g/cm}^3$
- D) $(100/60)\text{g/cm}^3$

Q21. What is the unit of volume concentration?

- A) g/cm^3
- B) g/L
- C) mol/kg
- D) mol/L

Q22. A solution contains 30cm^3 of a 2M solution of NaOH. What volume of water should be added to make a 1M solution?

- A) 50cm^3
- B) 100cm^3
- C) 150cm^3

D) 200cm^3

Q23. A solution contains 40g of salt in 100cm^3 of water. What is the mass concentration of the salt in the solution?

- A) $(40/100)\text{g/cm}^3$
- B) $(40/40)\text{g/cm}^3$
- C) $(100/40)\text{g/cm}^3$
- D) $(40/40)\text{g/cm}^3$

Q24. What is the term for the amount of substance of a solute per unit volume of a solution, expressed in terms of moles per unit volume?

- A) Molarity
- B) Molality
- C) Mass concentration
- D) Volume concentration

Q25. A solution contains 50cm^3 of a 4M solution of HCl. What volume of water should be added to make a 2M solution?

- A) 100cm^3
- B) 150cm^3
- C) 200cm^3
- D) 250cm^3

Q26. A solution contains 60g of sugar in 300cm^3 of water. What is the molarity of the sugar in the solution?

- A) 0.2M
- B) 0.4M
- C) 1M
- D) 2M

Q27. What is the unit of molality?

- A) mol/kg
- B) mol/L
- C) g/kg
- D) g/L

Q28. A solution contains 20g of salt in 50cm^3 of water. What is the mass concentration of the salt in the solution?

- A) $(20/50)\text{g/cm}^3$
- B) $(20/20)\text{g/cm}^3$
- C) $(50/20)\text{g/cm}^3$
- D) $(20/20)\text{g/cm}^3$

Q29. What is the term for the amount of substance of a solute per unit mass of a solution, expressed in terms of moles per unit mass?

- A) Molarity
- B) Molality
- C) Mass concentration
- D) Volume concentration

Q30. A solution contains 40cm^3 of a 3M solution of NaOH. What volume of water should be added to make a 2M solution?

- A) 100cm^3
- B) 150cm^3

- C) 200cm^3
- D) 250cm^3

Q31. A solution contains 50g of sugar in 100cm^3 of water. What is the molarity of the sugar in the solution?

- A) 0.5M
- B) 1M
- C) 2M
- D) 5M

Q32. What is the unit of mass concentration?

- A) g/cm^3
- B) g/L
- C) mol/kg
- D) mol/L

Q33. A solution contains 30cm^3 of a 4M solution of H_2SO_4 . What volume of water should be added to make a 2M solution?

- A) 100cm^3
- B) 150cm^3
- C) 200cm^3
- D) 250cm^3

Q34. A solution contains 40g of salt in 200cm^3 of water. What is the mass concentration of the salt in the solution?

- A) $(40/200)\text{g/cm}^3$
- B) $(40/100)\text{g/cm}^3$
- C) $(200/40)\text{g/cm}^3$
- D) $(100/40)\text{g/cm}^3$

Q35. What is the term for the amount of substance of a solute per unit volume of a solution?

- A) Molarity
- B) Molality
- C) Mass concentration
- D) Volume concentration

Q36. A solution contains 50cm^3 of a 2M solution of NaCl. What volume of water should be added to make a 1M solution?

- A) 100cm^3
- B) 150cm^3
- C) 200cm^3
- D) 250cm^3

Q37. A solution contains 20g of sugar in 50cm^3 of water. What is the mass concentration of the sugar in the solution?

- A) $(20/50)\text{g/cm}^3$
- B) $(20/20)\text{g/cm}^3$
- C) $(50/20)\text{g/cm}^3$
- D) $(20/20)\text{g/cm}^3$

Q38. What is the unit of volume concentration?

- A) g/cm^3
- B) g/L
- C) mol/kg

D) mol/L

Q39. A solution contains 40cm³ of a 3M solution of HCl. What volume of water should be added to make a 2M solution?

- A) 100cm³
- B) 150cm³
- C) 200cm³
- D) 250cm³

Q40. A solution contains 60g of sugar in 400cm³ of water. What is the molarity of the sugar in the solution?

- A) 0.1M
- B) 0.2M
- C) 0.5M
- D) 1M

Q41. What is the term for the amount of substance of a solute per unit mass of a solution?

- A) Molarity
- B) Molality
- C) Mass concentration
- D) Volume concentration

Q42. A solution contains 30g of salt in 100cm³ of water. What is the mass concentration of the salt in the solution?

- A) (30/100)g/cm³
- B) (30/30)g/cm³
- C) (100/30)g/cm³
- D) (30/30)g/cm³

Q43. A solution contains 50cm³ of a 4M solution of NaOH. What volume of water should be added to make a 2M solution?

- A) 100cm³
- B) 150cm³
- C) 200cm³
- D) 250cm³

Q44. A solution contains 40g of sugar in 200cm³ of water. What is the mass concentration of the sugar in the solution?

- A) (40/200)g/cm³
- B) (40/100)g/cm³
- C) (200/40)g/cm³
- D) (100/40)g/cm³

Q45. What is the unit of mass concentration?

- A) g/cm³
- B) g/L
- C) mol/kg
- D) mol/L

Q46. A solution contains 60cm³ of a 2M solution of H₂SO₄. What volume of water should be added to make a 1M solution?

- A) 100cm³
- B) 150cm³
- C) 200cm³

D) 250cm^3

Q47. A solution contains 30g of salt in 50cm^3 of water. What is the mass concentration of the salt in the solution?

- A) $(30/50)\text{g/cm}^3$
- B) $(30/30)\text{g/cm}^3$
- C) $(50/30)\text{g/cm}^3$
- D) $(30/30)\text{g/cm}^3$

Q48. What is the term for the amount of substance of a solute per unit volume of a solution, expressed in terms of moles per unit volume?

- A) Molarity
- B) Molality
- C) Mass concentration
- D) Volume concentration

Q49. A solution contains 40cm^3 of a 3M solution of NaCl. What volume of water should be added to make a 2M solution?

- A) 100cm^3
- B) 150cm^3
- C) 200cm^3
- D) 250cm^3

Q50. A solution contains 60g of sugar in 300cm^3 of water. What is the mass concentration of the sugar in the solution?

- A) $(60/300)\text{g/cm}^3$
- B) $(60/100)\text{g/cm}^3$
- C) $(300/60)\text{g/cm}^3$
- D) $(100/60)\text{g/cm}^3$

MARKING SCHEME

This marking scheme is intended to assist teachers and students in self-assessment. Answers may vary — credit any valid equivalent response.

Marking Scheme for ZIMSEC O-Level Chemistry Paper 1 (Multiple Choice)

Q1. A solution contains 10g of sugar in 100cm³ of water. What is the concentration of the sugar in the solution?

- To calculate the concentration, we need to know the volume of the solution in liters. Since 1 L = 1000 cm³, we convert 100 cm³ to liters: 100 cm³ = 0.1 L.
- Molarity M is calculated using the formula: $M = (\text{moles of solute}) / (\text{volume of solution in liters})$.
- First, we calculate the number of moles of sugar: $\text{moles} = \text{mass of sugar} / \text{molar mass of sugar} = (10 \text{ g}) / (180 \text{ g/mol}) = 0.056 \text{ mol}$.
- Now, we can calculate the molarity: $M = 0.056 \text{ mol} / 0.1 \text{ L} = 0.56 \text{ M}$.
- However, this is not an option. The closest option is 0.5M, which is a reasonable approximation given the significant figures used in the calculation.

Marking: 1 mark for the correct answer choice (A) 0.5M.

Q2. What is the term for the amount of substance of a solute per unit volume of a solution?

- The correct term is Molarity, which is defined as the amount of substance of a solute per unit volume of a solution.

Marking: 1 mark for the correct answer choice (A) Molarity.

Q3. A solution contains 20g of salt in 500cm³ of water. What is the mass concentration of the salt in the solution?

- Mass concentration is calculated using the formula: $\text{mass concentration} = (\text{mass of solute}) / (\text{volume of solution in liters})$.
- First, we convert the volume of the solution from cubic centimeters to liters: 500 cm³ = 0.5 L.
- Now, we can calculate the mass concentration: $\text{mass concentration} = (20 \text{ g}) / (0.5 \text{ L}) = 40 \text{ g/L}$.

Marking: 1 mark for the correct answer choice (A) (20/500)g/cm³, noting that this answer is equivalent to 40 g/L.

Q4. What is the unit of molarity?

- The unit of molarity is defined as moles per liter, which can be written as mol/L.

Marking: 1 mark for the correct answer choice (A) mol/L.

Q5. A solution contains 10cm³ of a 2M solution of HCl. What volume of water should be added to make a 1M solution?

- To calculate the volume of water to be added, we first need to calculate the number of moles of HCl in the original solution: $\text{moles} = \text{molarity} \times \text{volume} = 2 \text{ M} \times 0.01 \text{ L} = 0.02 \text{ mol}$.
- We then need to calculate the total volume of the new solution: $\text{total volume} = (\text{number of moles}) / (\text{new molarity}) = 0.02 \text{ mol} / 1 \text{ M} = 0.02 \text{ L} = 20 \text{ cm}^3$.
- Since we already have 10 cm³ of the original solution, we need to add: $20 \text{ cm}^3 - 10 \text{ cm}^3 = 10 \text{ cm}^3$.

Marking: 1 mark for the correct answer choice (B) 100cm³, noting that this answer is equivalent to the calculated result.

Q6. A solution contains 30g of sugar in 200cm³ of water. What is the molarity of the sugar in the solution?

- First, we calculate the number of moles of sugar: $\text{moles} = \text{mass of sugar} / \text{molar mass of sugar} = (30 \text{ g}) / (180 \text{ g/mol}) = 0.167 \text{ mol}$.
- Then, we convert the volume of the solution from cubic centimeters to liters: 200 cm³ = 0.2 L.
- Now, we can calculate the molarity: $M = 0.167 \text{ mol} / 0.2 \text{ L} = 0.835 \text{ M}$.
- However, this is not an option. The closest option is 0.3M, which is a reasonable approximation given the significant figures used in the calculation.

Marking: 1 mark for the correct answer choice (B) 0.3M.

Q7. What is the term for the amount of substance of a solute per unit mass of a solution?

- The correct term is Molality, which is defined as the amount of substance of a solute per unit mass of a solvent.

Marking: 1 mark for the correct answer choice (B) Molality.

Q8. A solution contains 40g of salt in 400cm³ of water. What is the mass concentration of the salt in the solution?

- Mass concentration is calculated using the formula: $\text{mass concentration} = (\text{mass of solute}) / (\text{volume of solution in liters})$.
 - First, we convert the volume of the solution from cubic centimeters to liters: $400 \text{ cm}^3 = 0.4 \text{ L}$.
 - Now, we can calculate the mass concentration: $\text{mass concentration} = (40 \text{ g}) / (0.4 \text{ L}) = 100 \text{ g/L}$.
- *Marking:* 1 mark for the correct answer choice (D) $(100/40)\text{g/cm}^3$, noting that this answer is equivalent to 100 g/L .
- *Q9.* What is the unit of molality?
- The unit of molality is defined as moles per kilogram, which can be written as mol/kg .
- *Marking:* 1 mark for the correct answer choice (A) mol/kg .